

Microplastic Removal System Design

Final Design Report

Prepared for

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By

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1. Executive Summary

Material Recovery Facilities (MRF) play an important role in the process of microplastic processing in wastewater. Current wastewater facilities are capable of collecting and processing large microplastic particles. While current filtration systems capture large debris, they typically lose efficiency at the 5 micrometers (μm) threshold, which allows millions of particles per cubic meter to get into the environment. Furthermore, the lack of real-time monitoring leaves MRF coordinators, who are responsible for maintaining operations in recovery facilities, unable to detect filtration failures or track contamination. This report proposes an integrated microplastic detection and extraction robotic system designed to close the 5-10 μm safety gap, while maintaining the high-fidelity resolution required to detect particles even smaller than 5 μm . This ensures a redundant safety margin that captures the invisible pollutants standard MRF operations overlook.

This system is installed at the final stage of the MRF water effluent line, centered around a 200 cubic meter holding tank capable of managing industrial flow rates up to 7 L/s. Within this tank, a swarm of magnetically responsive microrobots actively identifies and captures dispersed microplastics. A key innovation of this design is the shift from vision-based robotics to non-optical microfluidic microwave sensing. By measuring dielectric properties rather than relying on visual confirmation, the system utilizes an onboard machine learning classifier to identify the unique dielectric signatures of specific polymers, such as polystyrene (PS), polyethylene (PE), and polyethylene terephthalate (PET), achieving over 94% detection accuracy even in low-visibility conditions typical of MRF effluent.

Our design fulfills the diverse requirements of facility stakeholders. MRF operators receive a continuous, machine-learning-assisted data stream for real-time filtration feedback, while facility management gains the verifiable records necessary for regulatory compliance. Beyond the facility, local communities benefit from significantly improved water quality and ecosystem protection.

Based on current literature, the system targets a minimum removal efficiency of 83% for particles in the critical 5–10 μm range. Key design constraints remaining for development include optimizing the balance between microrobot battery life and sensing range, selecting sustainable materials for the robots, and ensuring seamless integration with existing mechanical infrastructure.

By transitioning from passive, infrequent manual sampling to an autonomous, 24/7 monitoring and extraction framework, this system provides MRFs with a scalable solution to mitigate microplastic discharge. This design not only addresses current environmental risks but also future-proofs facilities against tightening global discharge regulations.

2. Introduction

Material Recovery Facilities (MRF) play an important role in the recycling system by sorting, cleaning and preparing recyclables for processing. However during the operations to do so, shredding, crushing, washing, and agitation, causes microplastics to be released into both the

water and the air. While current filtration systems are able to collect larger pieces of microplastics, they are often unable to collect particles smaller than 5 micrometers (μm) [1], which are then released into the environment. The scale of this release in recycling water can reach millions of particles per cubic meter, with typical recycling water containing microplastics ranging from approximately 5.97×10^6 to $1.12 \times 10^8 \text{ MP m}^{-3}$ [2]. Currently, these facilities lack the ability to acquire real-time data on these particles or the mechanisms to remove them effectively.

This report argues that MRFs must move beyond simple filters to address the 5–10 μm microplastic gap, while also extending detection capabilities to sub-5 μm particles. By using a robotic system that actively senses and identifies different plastics, facilities can finally provide the real-time data needed to prevent environmental contamination. By integrating non-optical microwave sensors with a magnetically responsive robotic swarm, MRFs can achieve the real-time identification and verifiable discharge records necessary to meet tightening environmental regulations that traditional systems cannot address.

3. Literature Review

To understand the problems and effects of microplastic contamination and build a system that works, we performed research on three main things: where the plastic originates, why it is harmful, and why current techniques are not fully optimizing their removal. Our research shows that MRFs are significant contributors to the systemic release of microplastics, discharging millions of particles through wash-water streams that bypass standard filtration systems.

While the adverse health and environmental impacts of microplastics have been established, existing identification tools remain insufficient for industrial applications. Standard vision-based systems are unable to see clear plastic in dirty water, and laboratory methods are too time-intensive for live tracing. This literature review looks at how we can bridge that gap. By identifying the limitations of current technologies, this report justifies the development of a real-time detection and extraction system. This design addresses the critical 5-10 μm filtration gap while providing a safety margin that extends thresholds to particles smaller than 5 μm , capturing particulates typically missed by standard filtration. The following sections outline our findings and explain the technical rationale behind this design approach.

3.1. Detection & Perception Methods

Accurate detection of the microplastics is a critical factor for autonomous robotic systems for recycling processes. Traditional work such as Fourier transform infrared spectroscopy (FT-IR), Raman spectroscopy, and pyrolysis gas chromatography-mass spectroscopy (Py-GC/MS) are the most chemically reliable methods for confirming microplastic identity. However, these methods are different considerably in resolution, cost, and compatibility from field conditions. Newer tools such as automated scanning electron microscopy with energy-dispersive X-ray spectroscopy (SEM-EDS) improve throughput but still fail to meet all needs for sensitivity, speed, cost, and portability simultaneously [1]. This limitation is particularly consequential for robotic applications, where real-time onboard processing is a practical necessity rather than a luxury.

Computer vision-based approaches offer a more viable pathway toward autonomous detection, but introduce their own failure modes. One of the most complete integrated perception-action systems in the current literature is an autonomous beach-surveying robot whose segmentation pipeline, built on thresholding and contour processing, struggles with particles that are buried beneath sand, share the color of their substrate, or are transparent [2]. The authors identify foundation models capable of real-time inference on embedded hardware as the most promising direction for improvement. This gap between current vision-based systems and the robustness required for real-world deployment represents one of the clearest opportunities in the field.

Beyond optical methods, recent work has explored non-destructive sensing modalities for specific contexts. Integrating microwave sensing with pattern recognition algorithms has shown promise for detecting polystyrene contamination in wheat flour, enabling rapid multi-task analysis without sample destruction [1]. Recent advancements have specifically targeted the challenges of aqueous environments through microfluidic integration. For instance, an advancement demonstrated a microwave sensor integrated with a microfluidic channel capable of real-time detection by measuring variations in reflected signals as particles pass through a sensing zone, achieving sensitivity even in high-flow conditions. This is complemented by the development of portable, multichannel measurement systems that can simultaneously read multiple microwave sensors, allowing for the characterization of various particle sizes and polymer properties in a compact form factor suitable for robotic integration [3][4].

Further research has pushed the boundaries of sensitivity and specific classification of microplastics. Active differential microwave sensors have been successfully used to distinguish between different polymers, such as polystyrene (PS) and polyethylene (PE), in the 10–24 μm range with over 94% accuracy [5]. By coupling high-resolution resonators with machine learning-assisted classification, these platforms can overcome overlapping dielectric responses to provide both quantification and identification in a single device [6]. While feasibility studies acknowledge that background factors like temperature and salinity can influence resonance frequency shifts, exploratory work in planar resonators continues to refine the dielectric contrast between water (permittivity 80 farads/m) and microplastics (permittivity 2–3 farads/m) to ensure reliable in-situ monitoring [7].

While this approach is currently domain-specific, validated only for a single polymer type in a controlled food matrix, it suggests that hybrid sensing strategies combining multiple modalities could help address the limitations of any single technique. A broader survey of detection technologies similarly confirms that no unified method currently satisfies the full range of operational requirements across environmental monitoring contexts [8].

3.2. Robotic & Autonomous Removal Systems

Recent advancements in environmental surveillance have shifted toward the role of microplastic robotics to address the limitations of traditional monitoring. While microplastic robotics represent a significant shift in environmental surveillance, the literature reveals that their effectiveness is limited by significant perception gaps, a variety of active removal mechanisms that cannot be easily evaluated or compared, and a lack of continuous sampling protocols.

A recurring limitation in current robot platforms is the inability to distinguish microplastics from their surrounding environments. For instance, autonomous rovers designed for navigating a beach surveying often failed to detect particles that are transparent or covered by sand [2]. While laboratory tools like microwave sensing and pattern recognition have been successful in controlled settings, such as detecting a type of microplastic in wheat flour, translating this perception to complex, real-world environments remains a challenge [9][10]. Even advanced tools like FT-IR and Raman spectroscopy, while reliable, are often too slow or expensive for real-time processing used by an autonomous system.

Furthermore, the literature presents a wide variety of active removal mechanisms, ranging from large-scale autonomous rovers to microscopic bacterial swarms, making it difficult for researchers to establish a single 'best' method. On one end of the spectrum, researchers are exploring nature-inspired bacterial microrobots that use swarming motions to capture particles. On the other hand, magnetic nanorobots are proposed as a chemical-free remediation tool [11][12]. However, as noted in the review of magnetic systems, a lack of standardized testing protocols prevents these different technologies from being compared under realistic operating conditions.

Finally, the effectiveness of these robotic systems is delayed by the absence of continuous sampling protocols. Most current research identifies recycling facilities as major sources of pollution, yet data collection remains episodic and manual [13], [14], [15]. Since tests only happen at set times, researchers can't see the slow drop in filter quality that happens between those checks. This measurement gap justifies the development of robotic systems that move beyond experimental prototypes toward 'deployment-grade' sensors capable of 24/7 monitoring.

3.3. Sources, Spread & Environmental Impact

Although recycling is widely promoted as a key strategy to reduce plastic pollution and move toward a circular economy, recent research highlights that plastic recycling facilities themselves can be a distinct source of microplastic pollution. The mechanical processes involved in plastic recycling, particularly shredding, washing, and pelletizing generate abundant microplastic fragments as larger plastic items are broken down into smaller pieces that are difficult to fully contain or capture [1]. Pilot studies of mixed plastics recycling facilities have measured microplastic concentrations between 5.97×10^6 and 1.12×10^8 particles per cubic meter in wash water, indicating that the plastic preparation and cleaning stages can produce significant particulate waste streams even before final recycling outputs are created [1]. Conventional filtration systems used at such facilities tend to remove larger particles effectively, but very small microplastics ($<5 \mu\text{m}$) pass through treatments and are discharged into receiving waters, potentially equating to tens to over a thousand tonnes of microplastics released annually from a single plant's effluent if mitigation measures are not optimized [1]. In addition to waterborne emissions, microplastic particles can become airborne through dust and handling losses within facilities, worsening the overall environmental effects of this issue.

3.4. Human Health & Ecosystem Effects

Recycling facility wash water can be a high intensity point source of microplastics to receiving waters. The smallest portions (microplastics below 5 micrometers), are most likely to bypass

filtration and be undercounted, making them the highest value target for microrobotic effluent capture and monitoring. A pilot UK recycling facility study estimated wash-water concentrations of around 5.97×10^6 to 1.12×10^8 particles m^3 and found that filtration removed most particles above 5 micrometers but generally could not remove particles under 5 micrometers effectively [1]. Wastewater is widely recognized as a transport method from land-based sources to aquatic ecosystems, and are commonly described as source-to-sea risks including organism ingestion and physical harm, with effects varying by particle size, shape, and chemical makeup.

Human exposure occurs primarily via ingestion and inhalation. The WHO notes that microplastics can enter drinking water sources via runoff and industrial effluent, but that risk characterization remains limited by incomplete exposure and various data, especially for smaller particles.

3.5. Discussion

Our team is developing a microrobotic swarm designed to enhance microplastic removal at the final stage of the wastewater treatment process. The system is installed within a holding tank along the water effluent line of a MRF, where it deploys microrobots to actively collect contaminants, followed by magnetic retrieval and reuse. Additionally, we are designing the robots to travel through connecting pipes to expand contamination coverage beyond the holding tank when necessary.

A key design choice of ours was the use of a distributed swarm of small microrobots rather than a single large filtration mechanism. We decided on this swarm approach in order to increase total surface area coverage within the tank, improve collection efficiency, and provide redundancy in case individual robots fail. The feasibility of a swarm approach is supported by experimental data from magnetic microrobot swarms, demonstrating 83% to 89% microplastic removal in these settings [11][16]. These microrobots capture dispersed microplastics using magnetic fields. This model also allows for scalability, as the number of robots working together can be adjusted based on the size and contamination levels. Each microrobot will be magnetically responsive, enabling the holding tank system to deploy magnetic microrobots to collect micro-plastics and use magnets to retrieve them, where collected microplastics will be separated and the robots cleaned for reuse. This system reduces operational waste and supports long-term cost efficiency.

Another key design choice was incorporating pipe navigation capability. In addition to operating within the holding tank, the robots are designed to travel through connected pipes in order to collect microplastics that accumulate before reaching the tank. The robots will have a compact geometry to prevent clogging and ensure safe passage through curved or narrow pipe sections.

Furthermore, our team is still evaluating the specific material that will be used for the hydrophobic capture surface. Certain hydrophobic capture surfaces have chemical or polymer coatings, which could affect both the amount of microplastics collection and environment safety [11][17]. While chemical coatings allow more precise polymer attraction, polymer coatings provide a higher collection density. In addition to these methods, our team is still evaluating a way to retrieve these microplastics without damaging the robot, which will hopefully be resolved around the time of our Whiteboard Downselect Talk.

4. Stakeholder Analysis

Our implementation of a microplastic detection and removal system depends on understanding the needs of those who operate, manage, and are affected by the facility's system and environmental output. Our analysis categorizes stakeholders into three groups based on their level of involvement and the directness of the system's impact. By identifying the specific "gaps" each group faces, ranging from the daily operational needs of workers on the line to the long-term compliance goals of management and the health concerns of local residents, we can ensure our design provides practical value. The following sections outline how our real-time sensing technology addresses these needs, moving the facility from a reliance on infrequent manual testing to a state of continuous, data-driven environmental management.

4.1. Primary Stakeholder

The MRF operators are identified as the primary stakeholders for this project. Those operators refer to the individuals who are directly affected by the recycling problems while they are responsible for implementing the solution. They also oversee the recycling process such as crushing, washing, and shredding stages. Since they are in charge of the physical infrastructure and daily workflow, they are the major stakeholders who are easier to deploy the autonomous system for microplastic operation due to their direct access to the facilities.

One significant gap between what MRF operators need and what they have is the operator needs to know whether the current filtration and containment system are functioning adequately in real time. However, these feedback are currently heavily relied on infrequent manual sampling that may not capture the dynamic continuous nature of microplastic release when performing active processing [13]. This gap indicates that the filter failure or spikes in particle release may be undetected, exposing workers and the surrounding environment to unnecessary contamination. In addition, operators face a compliance gap where environmental regulations about microplastic discharge are currently tightening but recycling facilities lack the monitoring infrastructure to demonstrate continuous regulatory compliance. This means that providing operators with autonomous, real-time monitoring data stands a better chance of closing these gaps and enables operators to take corrective action immediately, documenting the facilities' environmental performance.

4.2. Secondary Stakeholder

We have identified Facility Management and Environmental Compliance Officers as secondary stakeholders because they are responsible for the facility's adherence to tightening environmental regulations and the overall economic efficiency of the plant. They serve as the decision-makers who must validate that the removal system meets legal discharge permits and internal sustainability goals.

There is currently a significant gap between the management's need for verifiable, continuous data and the data they currently receive through occasional manual testing. Because manual samples are only checked every few hours or days, management often lacks the data necessary to identify when a primary filtration unit begins to lose efficiency before a total failure occurs [13]. This lack of data creates a high-risk limitation where facility leaders may unintentionally report

inaccurate discharge levels simply because traditional methods fail to detect transparent or $< 5\text{-}10\text{ }\mu\text{m}$ particles.

Our system solves this by providing a 24/7 digital record of exactly what types of plastic are leaving the facility. This allows management to fix problems before they become a crisis [14]. By using these reliable sensors, the facility can avoid expensive government fines and keep its environmental certifications.

4.3. Tertiary Stakeholders

In a system designed to detect and remove microplastics from the water runoff of a MRF, several groups are considered tertiary stakeholders because they are affected by the outcomes of the system but are not directly involved in its operation or management. Local communities and residents fall into this category because they may experience environmental or health benefits if microplastics in wastewater are reduced, such as improved local water quality or reduced pollution in nearby waterways. However, they do not typically participate in the design, implementation, or daily operation of the treatment system. Their role is largely indirect, consisting of receiving the environmental impacts of the facility and potentially advocating for better environmental practices.

Similarly, environmental non-governmental organizations (NGOs), industry associations and trade groups, and media or news organizations are tertiary stakeholders because they influence public perception, policy discussions, and industry norms without directly managing the microplastic removal system. Environmental NGOs may monitor pollution levels, advocate for stricter environmental protections, or raise awareness about microplastic contamination, but they are not responsible for operating treatment technology at the MRF [18]. Industry associations and trade groups may disseminate best practices or encourage members to adopt new technologies, yet they typically do not implement the system themselves. Meanwhile, media and news outlets play an informational role by reporting on environmental issues and technological solutions, shaping public awareness and discourse. While these groups can indirectly influence how widely such systems are adopted, their involvement remains external to the core functioning of the microplastic removal process, making them tertiary stakeholders.

4.4. Requirements

Table 1 highlights the operational requirements and specifications necessary for the system to effectively help with microplastic removal at a MRF. These criteria prioritize the transition from passive filtration to an active, "intelligent" monitoring and removal system. The specifications are centered on the dielectric sensing capabilities, which allow for high-accuracy detection and polymer identification in environments where optical systems fail. By bridging the $5\text{--}10\text{ }\mu\text{m}$ detection gap and providing real-time concentration data, these requirements ensure the system not only captures microplastics but also provides the diagnostic feedback necessary for facility operators to maintain regulatory maintenance and trace mechanical failures.

Table. 1: Project Design Requirements and Specifications

Requirements	Specifications
The system must identify microplastics regardless of color, transparency, or water turbidity.	Achieve a detection accuracy of $\geq 94\%$ for particles in high-turbidity environments using dielectric resonance [5].
The system must categorize particles by specific polymer types to assist in facility leak tracing.	Distinguish between at least three polymer types (e.g., PS, PE, PET) using ML-assisted frequency sweeps [6].
The system must produce continuous concentration data compatible with health risk assessment models	Provide a live data feed of particle counts per cubic meter to MRF operators [9].
The primary deployment mechanism must physically remove identified particles from the effluent.	Achieve $83\% \leq$ removal efficiency using a magnetically responsive retrieval framework [11].
The treatment unit must contain daily effluent without disrupting facility operations.	200 m ³ holding tank capacity designed for peak flow of 7 L/s or 25 m ³ /hr [13][19][20][21].
The system must capture microplastics that bypass standard mechanical filtration.	Capable of detecting and capturing particles in the 5-10 μm range to create a redundant safety margin and capture sub-5 μm particles typically missed by mechanical filtration [13].

Current MRF filtration systems are prominent for catching large debris, but miss the tiny particles $< 5\mu\text{m}$ [22]. Our system is designed to catch everything up to $10\mu\text{m}$, by targeting the larger size, we create a larger safety margin that overlaps with existing filters. Our robots catch the highest concentration of microplastics and prevent any particles missed by the first set of filters from escaping into the water.

Our team had to balance the need for high-accuracy sensing with the practical demands of a fast-moving recycling facility. While the community and environment groups (Tertiary Stakeholders) want 100% of microplastics removed, the facility operators (Primary Stakeholders) cannot have a system that slows down their 7 L/s water flow. By engineering a system capable of detecting particles even smaller than 5 μm , we established a redundant safety margin that exceeds basic requirements. With this, we decided that achieving 94% accuracy was the most important goal because it gives management the verifiable data they need for regulations, while a 83% removal rate ensures the system catches a significant amount of plastic without disrupting the facility's daily operations.

4.5. Connections

The needs of the primary, secondary, and tertiary stakeholders are very closely connected through the shared goal of reducing microplastic pollution from recycling facilities. MRF operators serve as the primary stakeholders as they directly manage the infrastructure where microplastics are generated and released. Their need for real-time monitoring and filtration performance data directly impacts researchers, who rely on accurate and consistent datasets in order to study microplastic release patterns, evaluate mitigation strategies, and develop new technologies. By providing sensing and removal capabilities, the proposed system helps both operators and researchers address the shortcomings of manual sampling and inconsistent measurement practices.

The benefits of improved monitoring also extends to various tertiary stakeholders, such as local communities and their residents, environmental organizations, and the media. Communities living near recycling facilities are also directly impacted by environmental contamination, making reliable pollution monitoring crucial for maintaining safety and protecting local ecosystems and public health. Media organizations help communicate environmental risks and technological solutions to the public, helping increase awareness and accountability. By enabling continuous monitoring and extraction/removal of microplastics, the proposed system creates value shared across all stakeholder categories by improving environmental transparency, regulatory compliance, and more general trust in the recycling system.

5. Design Description & Rationale

Our microplastic removal system is designed to provide real-time detection in high-turbidity environments where traditional filtration and optical sensors fail. The design integrates a microrobot swarm with a microfluidic microwave sensor within a 200 m³ holding tank at the final stage of the MRF's water effluent line. Each component was chosen to solve the specific limitations found in our research, specifically, the inability of standard cameras to identify transparent or sub-5 micrometer particles. By focusing on these technical gaps, the system provides the continuous monitoring mechanism that current industrial facilities lack.

5.1. Objectives

The primary objective of this design is to sense and remove microplastics from wastewater within an MRF. However due to the size of the effort, this report will outline the overall system and assume those parts are already well engineered and then focus on one small part of the system. We will focus on the sensing of microplastics in waste water and the reason for which will be outlined later. Our objective with this sensing is that it must identify microplastics and their specific polymer types, regardless of its color, transparency or the water it is.

5.2. Design Requirements

The overall system needs to comply with requirements specified in the stakeholder analysis, which all fall into the following categories: Concentration Data, Deployment, Collection, and Sensing. For Concentration Data, the system has to provide continuous data about the concentration of the microplastics in water to some system that workers can access. For

Deployment, the system has to both release and collect the microrobots from the water tank with a high enough efficiency to not add to the current problem. Collection entails, both a large enough tank to collect all effluent and a large enough intake pipe to not slow down normal operation. Sensing, which is what we will focus on for the majority of this report, includes both identifying microplastics to a high accuracy regardless of color transparency or the water's turbidity in addition to identifying the specific type of polymer that the microplastic is.

5.3. Overall System Design

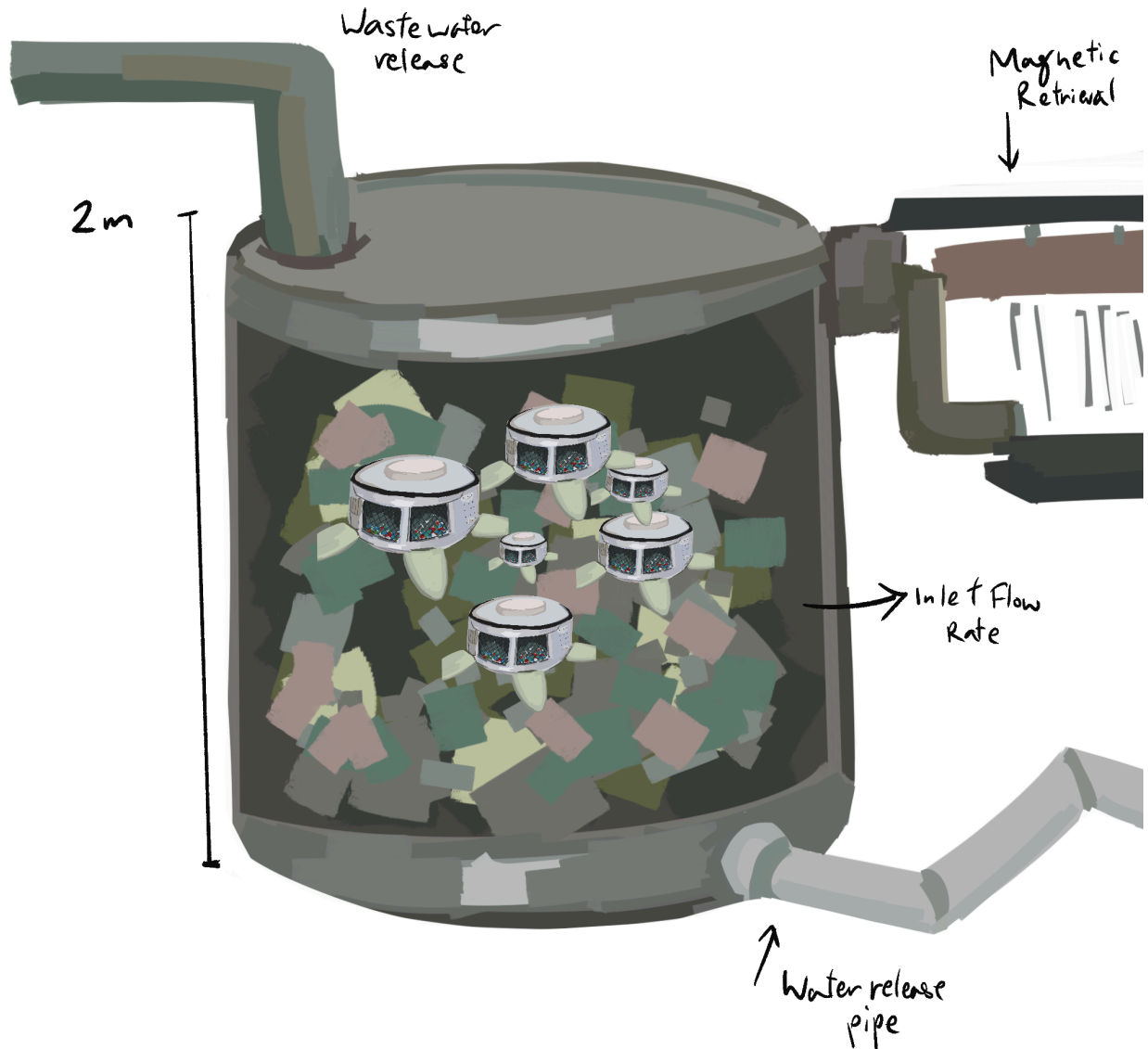


Fig. 1: Overview of the microplastic recycling system using microrobot swarm.

A 200 m³ holding tank serves as the central environment for the sensing and capture process, integrated directly into the MRF's final water effluent line (Fig. 1). This tank is engineered to

manage a peak flow of 7 L/s or 25 m³ /hr, so that this matches standard industrial washing and shredding cycles, ensuring our system does not cause mechanical backups or interfere with the facility's primary operations [19][21][23]. By treating the water on-site, we address the "shipping cost" burden identified in our research. The facility avoids the massive expense of trucking contaminated "sludge" or wastewater to external hazardous waste sites.

5.4. Design Downselection Prioritizing Sensing of Microplastics

During the initial phase of this project, our team evaluated the overall system design. However, after reviewing current literature, stakeholder needs, and the documented failure rates of vision-based robotics in turbid water, we decided to downselect our focus to the sensing and classification capabilities of the robot. (Detailed schematics of the initial design and the discarded optical prototypes are provided in Appendix A). We identified that the primary technical barrier in microplastic remediation stems from the inability to accurately identify particles in real-time. By prioritizing the development of the microfluidic microwave sensor, we ensure the system provides the high-accuracy data (> 94%) required for effective regulatory leak tracing prior to the large-scale mechanical implementation of the full system. This strategic focus allowed us to engineer a robust, non-optical solution that remains functional in environments where traditional systems fail.

5.5. Microfluidic Microwave Sensor

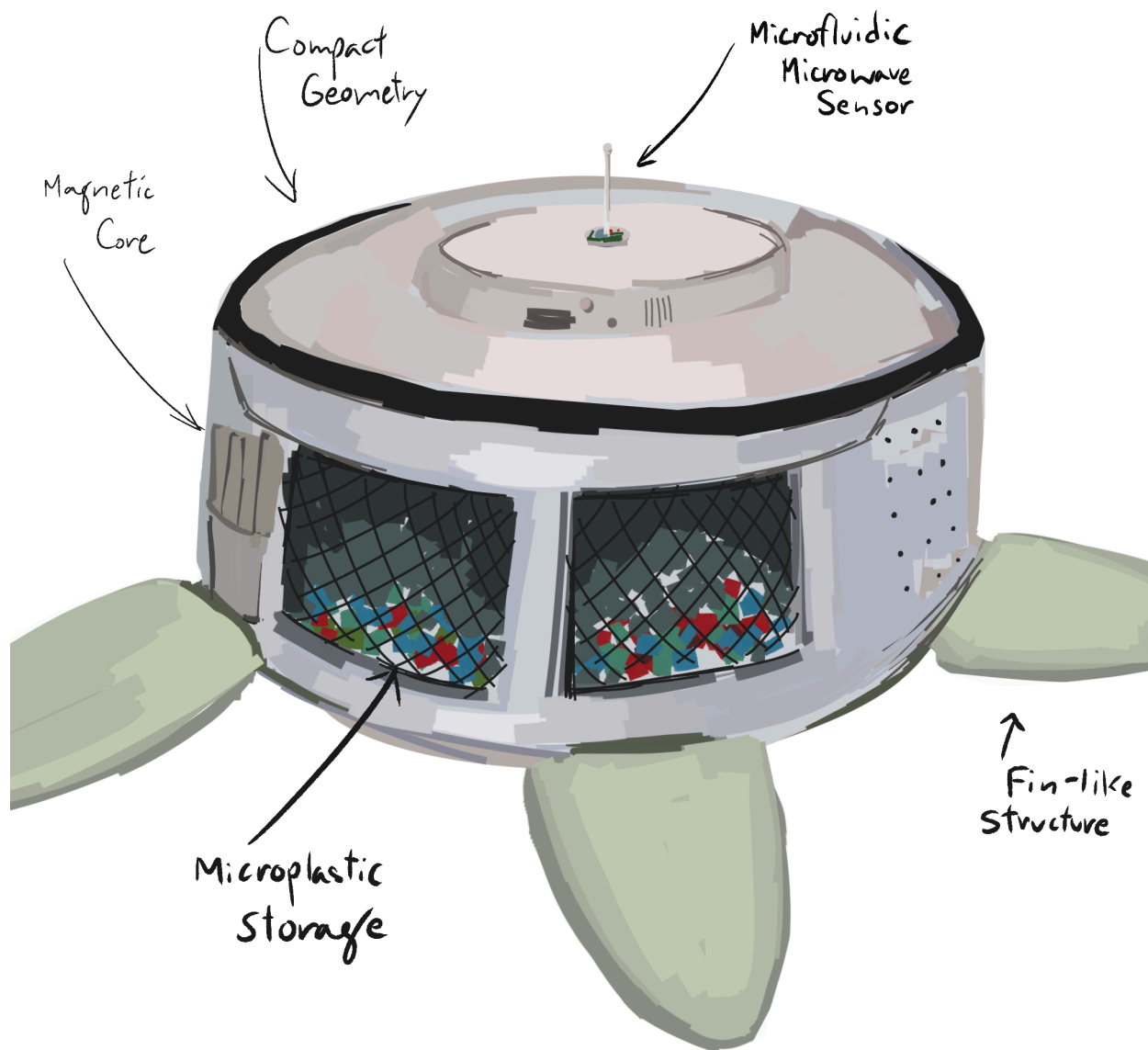


Fig. 2: Detailed sketch of the microbot used to collect the microplastics.

As shown above (Figure 2), the core innovation of each robotic unit is a non-optical, microfluidic microwave sensor. This replaces traditional cameras, which our Literature Review proved are ineffective in murky water [2]. The sensor measures Relative Permittivity and dielectric constants, so that the sensor can "see" particles even in total darkness or turbid (cloudy) wastewater. This allows for a $> 94\%$ detection accuracy [5].

A dedicated decision-making pipeline translates these raw electromagnetic signals into actionable data for facility stakeholders (Fig. 3). As a particle enters the microfluidic channel, the

system initiates high-speed frequency sweeps to capture a unique dielectric signature. This reasoning process allows for real-time identification without the delays associated with lab testing.

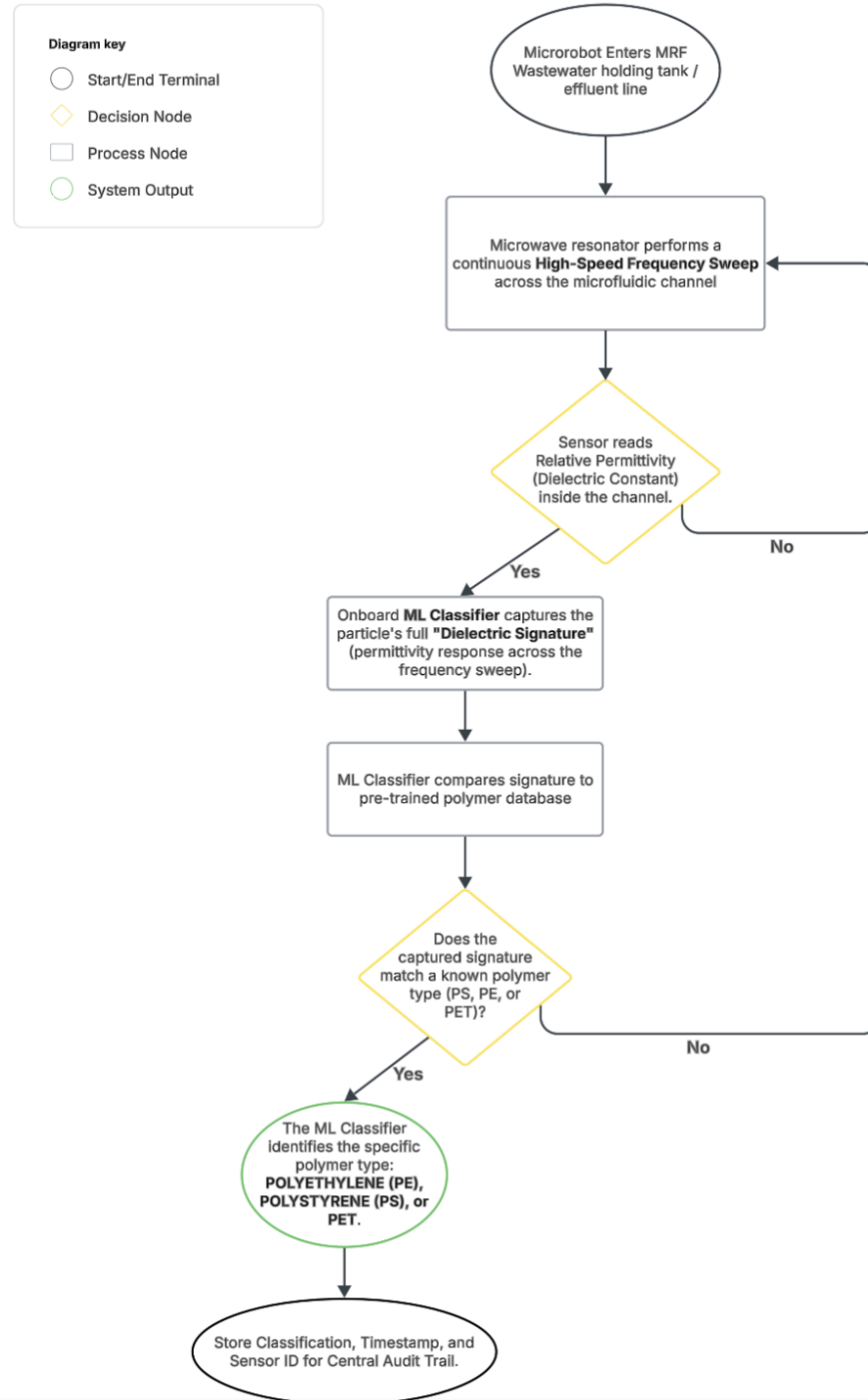


Fig. 3: Decision logic for the onboard machine learning classifier during particle identification.

The onboard Machine Learning (ML) classifier compares these signatures against a pre-trained library to distinguish between specific polymers like PS, PE, and PET. By automating this classification, the system can differentiate between chemically similar plastics, such as polystyrene and polyethylene, with high precision [6]. This provides the "digital audit trail" required by facility management and environmental compliance officers to trace contamination spikes back to specific upstream machinery.

5.6. Cost-Effectiveness

Material Recovery Facilities (MRFs) are currently investing in high-end microplastic filtration systems that can cost up to \$500,000 [24], while overall facility renovations often reach into the millions of dollars [26]. This suggests that a solution priced below several million dollars would likely be considered cost-effective, while systems closer to \$500,000 would fall within the most competitive range.

To estimate the cost of the proposed solution, each component must be evaluated individually, as summarized in the table below (Figure 4). The primary containment tank is estimated to cost between \$12,000 and \$30,000 [26]. The microbots themselves are projected to cost between \$0.01 and \$5.00 per unit, with additional data-reading equipment costing approximately \$3,000 [27]. Based on a required concentration of 0.75 mg/mL [16], the total cost of microbots is estimated at approximately \$1,000,000, assuming a unit cost closer to \$0.10, which is considered a reasonable estimate. The collection system, which includes permanent magnets, is expected to cost approximately \$500 depending on implementation [28]. Finally, miscellaneous site-specific expenses are accounted for with an estimated \$20,000 allocation. Altogether, these components result in a total estimated cost of \$1,053,500. While this exceeds the cost of traditional filtration systems (~\$500,000), the added functionality of real-time microplastic sensing justifies the increase. Furthermore, the total remains within the broader spending range that MRFs have demonstrated willingness to invest in infrastructure and equipment.

In addition to the initial capital cost of the system, operational and maintenance costs must also be considered when evaluating overall feasibility. While much of the maintenance required will depend on site-specific conditions and operations context, two consistent cost factors can be expected: additional labor hours for system operation and oversight, and periodic microbot replacement or repair. However, without real-time system testing or pilot-scale implementation, it is difficult to accurately quantify the magnitude of these costs. Despite this uncertainty, both factors are expected to contribute meaningfully to the overall operational expense and should therefore be included in any comprehensive cost analysis.

Table 2: Breakdown of component pricing

Items	Cost
Water Tank	\$30,000
Microboots	\$1,000,000
Reading Equipment	\$3,000
Collection	\$500
Odds and Ends	\$20,000
Total	\$1,053,500

5.7. Constraints

The design of the microrobotics microplastic removal system is constrained by several technical, environmental, and operational limitations. These limitations vary greatly in complexity and potential solutions. First of all, the system must operate efficiently in high-turbidity wastewater environments, where optical sensing methods are more prone to failure due to low visibility and the presence of transparent particles [9]. This necessitates the use of non-optical sensing methods such as the aforementioned microfluidic microwave sensor, which can operate more reliably in murky conditions.

Second, the system must integrate seamlessly into existing Material Recovery Facility (MRF) infrastructure without causing disruptions to flow rates or requiring significant facility redesign [13]. The designated holding tank must also accommodate peak effluent flow rates of about 7 L/s while avoiding congestion or operational downtime.

Another key constraint is the particle size, as conventional filtration systems often fail to capture microplastics below 5 μm , allowing significant quantities to pass into wastewater streams [13]. Therefore, the system must reliably detect and remove particles from the 5 - 10 μm range to provide redundancy and improved environmental protection. On the logistical level, the system must maintain cost-effectiveness and reusability, requiring the microrobots to be easily retrievable, durable, and resistant to chemicals used in wastewater environments that may cause degradation or malfunctions [11][16].

Lastly, environmental safety constraints require that any materials used must not introduce additional contaminants into the MRF system [17].

5.8. Core Aspects of Robotics

The proposed system incorporates multiple fundamental aspects of Robotics, including Perception, Reasoning, and Actuation.

Perception

The system uses a microfluidic microwave sensor to detect microplastics based on dielectric properties instead of visual sensing. This allows for reliable detection within wastewater environments where traditional computer vision systems may fail from environment noise and signal loss [9].

Reasoning

An onboard machine learning classifier that analyzes frequency response data in real-time. This reasoning layer identifies specific polymer types, supporting autonomous decision-making for targeted removal and providing stakeholders with precise leak-tracing data.

Actuation

Physical interaction with the environment is achieved through magnetically controlled microrobots. These units use hydrophobic surfaces to capture microplastics, while external magnetic fields provide the actuation required to navigate and retrieve the robots once collection is complete [11][16].

5.9. Previous Research and Existing Solutions

Existing research on microplastic detection and removal spans a wide range of sensing and remediation techniques. This can range from visually looking for microplastics, chemical testing, microwaves and more [10]. Traditional detection methods such as Fourier-transform infrared spectroscopy (FT-IR), Raman spectroscopy, and pyrolysis gas chromatography-mass spectroscopy provide high chemical accuracy but are severely limited by high cost, slow processing time, and lack of compatibility with real-time field deployment [8].

More recent approaches have explored the use of computer vision based systems, but these methods struggle in real-world environments where microplastics may be transparent, hidden, or visually indistinguishable from their surroundings. For example, autonomous robotic systems designed for environmental monitoring have demonstrated failure cases when particles are buried or visually camouflaged [2]. This limits their general effectiveness in wastewater environments.

Alternative sensing techniques, more specifically microwave-based dielectric sensing, have demonstrated strong potential for real-time detection in complex environments. These systems can detect microplastics based on differences in permittivity between water and plastic materials, enabling non-optical identification even in murky environments [9]. Recent work has also shown that these sensors can achieve high classification accuracy when paired with machine learning models [9].

In terms of microplastic removal, existing solutions range from large-scale filtration systems to magnetic nanorobots. Although these approaches show promising removal efficiencies, many remain limited to laboratory environments and lack standardized evaluation methods.

Overall, current solutions either lack real-time capability, malfunction in complex environments, or are not easily deployable within existing industrial facility infrastructure. This gap motivates

the development of an integrated solution combining real-time sensing with compatibility and autonomous robot removal.

5.10. Levels of automation vs. Human Decision Making and Interaction

The system follows a human-supervised automation model, where core sensing and data collection processes are automated, while high-level decision making remains under human control.

The perception and classification components operate autonomously, continuously collecting and analyzing microplastic concentration data in real time. This allows the system to act on its own and carry out core functions such as detecting contamination trends and identifying polymer types without human intervention.

However, system-level decisions including initiating cleaning cycles, adjusting operating parameters, or responding to abnormal readings are all managed by human operators to maintain safety and proper deployment. This approach ensures flexibility, safety, and accountability, especially in an industrial environment.

This hybrid model balances the efficiency of automation while maintaining safety measures with the reliability and oversight of a human operator, reducing the risk of system malfunction and minimizing operator workload simultaneously.

5.11. Potential Translation of elements

Microbots designed to sense and collect microplastics could be useful in many environments beyond MRFs. In wastewater treatment plants, microbots could help capture microplastics that pass through normal filtration systems. They could also be used in industrial settings where plastic dust or particles are produced during manufacturing, helping reduce pollution at the source. In agriculture, microbots could potentially remove plastic contamination from soil caused by degraded mulch films or fertilizers. Even inside homes and buildings, they could be used in filtration systems to stop microplastics from ending up in drinking water, or microplastics from being output from residential neighborhoods as well. All of this in addition to simply releasing the microbots into oceans, rivers, and lakes to collect microplastics, which is what they are currently being designed for in our literature.

6. Conclusion

This project report proposed an integrated design of a real-time microplastic detection and extraction system that is aimed to be implemented in the final stage of the MRF. Driven by the fact that traditional filtration systems fail to capture microplastics less than 5 micrometers in size, our system leverages a magnetically controlled swarm of microrobots. Each microbot is deployed with a microfluidic microwave sensing subsystem to enable microplastics detection and collection in real time. The swarm we designed achieved up to 83% removal efficiency. The microwave-based dielectric sensors are capable of achieving reliable detection under turbid and low-visibility environments in the effluent tank where the traditional optic approaches do not work.

The proposed system provides measurable value across all the needs of stakeholders by replacing delayed manual testing with continuous, data-driven oversight. The MRF operator, the primary stakeholder of our system, obtains real-time feedback to detect filtration failure as they occur, while secondary management receives the verifiable data required for environmental compliance and internal sustainable goals. Furthermore, local communities benefit from the significant reduction of microplastic discharge into receiving waterways, ensuring the facility meets its environmental obligations.

6.1. Limitations

Several limitations of our current system exist. To begin with, materials selection for the microrobot capture surface has not been finalized. Both hydrophobic and polymer coating have trade-offs between material collection specifications and environmental safety. The optimal choice depends on additional toxicological and performance evaluation. Secondly, balance between sensing capabilities and onboard powering remain a significant engineering challenge. Third, even though the microwave sensing achieves over 94% polymer classification accuracy in controlled laboratory environments, the performance where the microbots are operating under variable salinity, temperature, and dynamic flow conditions of the wastewater of real MRF effluent has not been fully tested and validated.

6.2. Potential Next Steps

The priority of our future work is resolving the material selection of our microbots through comparative testing of candidate hydrophobic coating for both collection efficiency and chemical safety in aqueous environments. Since our current design is based on theoretical assumptions, prototypes of subsystems including onboard sensors, magnetic controller unit, and microplastic extraction mechanism should be tested and validated.

On the system integration side, a small-scale pilot deployment in a controlled tank environment setting can better evaluate the swarm coordination, magnetic retrieval, and robot cleaning cycle end-to-end before the full-scale MRF installation. With such standardized testing protocol for removal efficiencies, consistency with emerging literature recommendations, would also allow the system performance to be compared against other approaches. In the long term, we aim to modularize our system with actuation architecture positions to allow deployment beyond MRF into wastewater treatment facilities, industrial manufacturing supply chain, and residential filtration applications, which expands the potential impact of our core design.

References

- [1] M. Sunil et al., “Exploring the Detrimental Effects of Nanoplastics on Ecosystems and Human Health,” *J. Hazard. Mater. Adv.*, vol. 19, p. 100831, Aug. 2025.
- [2] H. Iqbal, K. Rex, J. Shirley, C. Baiz, and C. Claudel, “A Novel Autonomous Microplastics Surveying Robot for Beach Environments,” 2025, arXiv:2508.02952.
- [3] A. Barrancos, V. Macedo, P. M. Ramos, and L. S. Rosado, “Microplastics Detection with Microfluidics Integrated with a Microwave Sensor,” *Meas. Sens.*, vol. 38, p. 101435, May 2025.
- [4] A. Barrancos, D. Rosinha, J. Assis, and L. S. Rosado, “Portable Multichannel Measurement System for Real-Time Microplastics Assessment Using Microwave Sensors,” *Sensors*, vol. 26, no. 2, p. 669, Jan. 2026.
- [5] Y. C. Alatas, U. Tefek, S. Salehin, H. Alhmoud, and M. S. Hanay, “Rapid Differentiation between Microplastic Particles Using Integrated Microwave Cytometry with 3D Electrodes,” *ACS Sens.*, vol. 10, no. 3, pp. 1729–1735, Mar. 2025.
- [6] M. I. O. Souza et al., “A Microwave-Based Sensing Platform for Microplastic Detection and Quantification: A Machine Learning-Assisted Approach,” *IEEE Sensors J.*, vol. 26, no. 7, pp. 10125–10133, Apr. 2026.
- [7] P. Zhao et al., “Detection of Microplastics by Microfluidic Microwave Sensing: An Exploratory Study,” *Sens. Actuators A Phys.*, vol. 383, p. 116154, Mar. 2025.
- [8] A. Kumar, M. Nehra, N. K. Singhal, and S. Kumar, “Recent Progress and Technological Advancements for Detection of Micro/Nano-plastics in the Environment,” *Adv. Colloid Interface Sci.*, vol. 351, p. 103817, May 2026.
- [9] J. Zhu, Z. Xu, and H. Jiang, “Integrating Microwave Sensing and Pattern Recognition Algorithms for Multi-Task Detection of Microplastics in Wheat Flour,” *Microchem. J.*, vol. 220, p. 116717, Jan. 2026.
- [10] S. Das et al., “Microplastic Contamination and Detection in Food Systems: A Review of Machine Learning, Traditional Methods, and Other Relevant Factors,” *Microchem. J.*, vol. 218, p. 115440, Nov. 2025.
- [11] S.-J. Song, J. Kim, R. Gabor, R. Zboril, and M. Pumera, “Magnetically Driven Living Microrobot Swarms for Aquatic Micro- and Nanoplastic Cleanup,” *ACS Nano*, vol. 19, no. 30, pp. 27259–27269, Aug. 2025.
- [12] S. Briceño, J. E. Arevalo-Fester, and I. A. Fierro-Sanchez, “Sustainable Magnetic Nanorobots for Microplastics Remediation,” *Microplastics*, vol. 4, no. 4, p. 97, Dec. 2025.
- [13] E. Brown, A. MacDonald, S. Allen, and D. Allen, “The Potential for a Plastic Recycling Facility to Release Microplastic Pollution and Possible Filtration Remediation Effectiveness,” *J. Hazard. Mater. Adv.*, vol. 10, p. 100309, May 2023.
- [14] G. Suzuki et al., “Mechanical Recycling of Plastic Waste as a Point Source of Microplastic Pollution,” *Environ. Pollut.*, vol. 303, p. 119114, Jun. 2022.
- [15] M. J. Stapleton, A. J. Ansari, A. Ahmed, and F. I. Hai, “Evaluating the Generation of Microplastics from an Unlikely Source: The Unintentional Consequence of the Current Plastic Recycling Process,” *Sci. Total Environ.*, vol. 902, p. 166090, Dec. 2023.

- [16] M. Ussia, M. Urso, C. M. Oral, X. Peng, and M. Pumera, “Magnetic Microrobot Swarms with Polymeric Hands Catching Bacteria and Microplastics in Water,” *ACS Nano*, vol. 18, no. 20, pp. 13171–13183, May 2024.
- [17] J. He et al., “Macroscopic and Microscopic Investigation of Adsorption Mechanisms of Phenanthrene and its Derivatives on Polyacrylonitrile Microplastics,” *Spectrochim. Acta A Mol. Biomol. Spectrosc.*, vol. 352, p. 127563, May 2026.
- [18] S. Islam, “The Role of NGOs and Advocacy Groups in Environmental Management,” ResearchGate, 2025. doi: 10.13140/RG.2.2.14614.05447.
- [19] R. L. Smith, S. Takkellapati, and R. C. Riegerix, “Recycling of Plastics in the United States: Plastic Material Flows and Polyethylene Terephthalate (PET) Recycling Processes,” *ACS Sustainable Chem. Eng.*, vol. 10, no. 6, pp. 2084–2096, Feb. 2022.
- [20] S. L. Bradshaw, H. A. Aguirre-Villegas, S. E. Boxman, and C. H. Benson, “Material Recovery Facilities (MRFs) in the United States: Operations, Revenue, and the Impact of Scale,” *Waste Manage.*, vol. 193, pp. 317–327, Feb. 2025.
- [21] M. N. Puteri, L. T. Gew, H. C. Ong, and L. C. Ming, “Technologies to Eliminate Microplastic from Water: Current Approaches and Future Prospects,” *Environ. Int.*, vol. 199, p. 109397, May 2025.
- [22] “Microplastics are everywhere — we need to understand how they affect human health,” *Nat. Med.*, vol. 30, no. 4, p. 913, Apr. 2024.
- [23] N.-B. Chang, E. Davila, B. Dyson, and R. Brown, “Optimal Design for Sustainable Development of a Material Recovery Facility in a Fast-Growing Urban Setting,” *Waste Manage.*, vol. 25, no. 8, pp. 833–846, Oct. 2005.
- [24] Ken Research, “Global Microplastics Removal Technologies Market | 2019 – 2030,” Apr. 14, 2026. [Online]. Available: <https://www.kenresearch.com/global-microplastics-removal-technologies-market>
- [25] Recycling Today, “MRF Retrofits: A Q&A with Nathaniel Egosi,” Apr. 14, 2026. [Online]. Available: <https://www.recyclingtoday.com/article/mrf-retrofit-egosi-qa/>
- [26] Alibaba, “304 316 Stainless Steel Storage Tank (200m3) for Food Grade Materials,” Apr. 14, 2026. [Online]. Available: https://www.alibaba.com/product-detail/304-316-Stainless-Steel-Food-Grade_1601164047226.html
- [27] M. Seggio et al., “Toward Nano- and Microplastic Sensors: Identification of Nano- and Microplastic Particles via Artificial Intelligence,” *ACS Omega*, vol. 9, no. 17, pp. 18984–18994, Apr. 2024.
- [28] totalElement, “1 Inch Powerful Neodymium Rare Earth Large Cylinder Magnet N52,” Apr. 14, 2026. [Online]. Available: <https://totalelement.com/products/totalelement-1-x-1-inch-neodymium-rare-earth-cylinder-magnet-n52>

Appendix

A

The following documentation outlines the initial, overall system architecture evaluated during the first phase of the design process. Initially, our team aimed to provide a broad solution to the entire mechanical lifecycle of our idea of having microrobots be a solution to sensing and collecting microplastics. However, as the technical complexity of reliable identification in turbid water became evident, the team strategically downselected the project scope to focus on the high-fidelity sensing and classification of polymers.

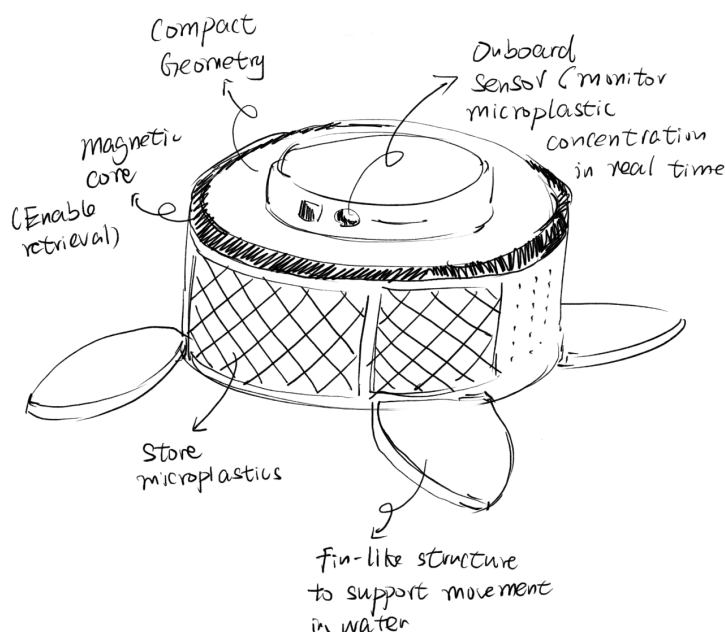


Figure A.1: Detailed sketch of the microbot used to collect the microplastics.

Early iterations of the system (see Fig. A.1) included an onboard optical camera for particle identification. This design was predicated on the assumption that edge-detection and contour-segmentation algorithms could effectively isolate plastic particulates from the surrounding aqueous medium. However, during the Design Rationale phase, this approach was formally discarded due to two primary physical constraints: the low refractive index contrast between transparent polymers and water, and the high turbidity of MRF effluent. The team concluded that relying on optical sensors in a murky environment would result in false-negative results, triggering the pivot to microwave-based dielectric sensing.